



ONDA LIFE

Technical Architecture & Product Ecosystem



PROJECT OVERVIEW • ENGINEERING DOCUMENTATION • 2026

The ONDA Ecosystem: Mobile App & Knowledge Platform

IOS & ANDROID

Mobile Application

-  **Biofeedback:** Real-time HRV practices and neural optimization.
-  **Gamification:** OND currency, levels, and digital artifacts.
-  **Personalization:** AI-driven emotional checks and tailored sessions.

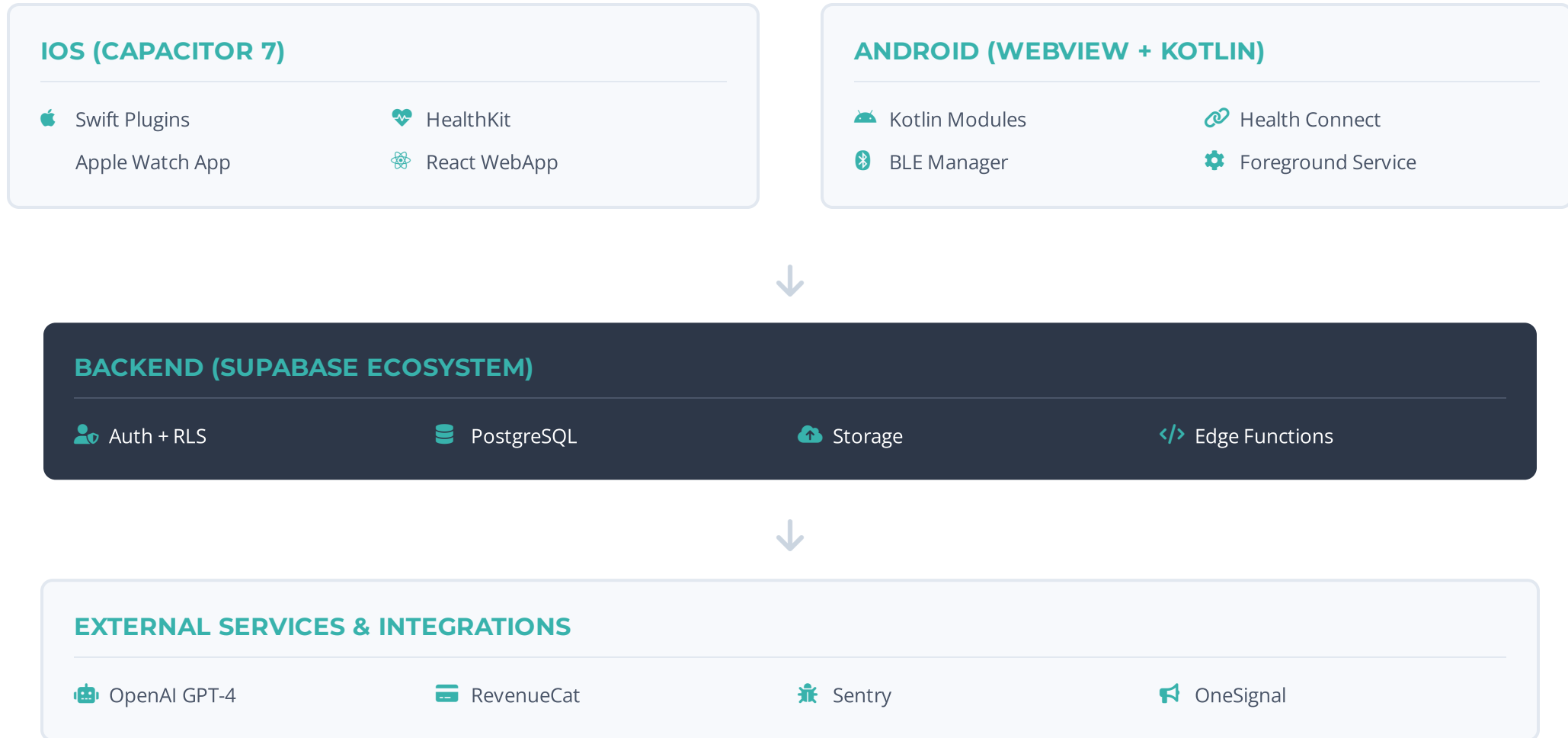
WEB / SEO

Landing & Knowledge Site

-  **Content Engine:** 68 articles and 216 glossary terms.
-  **SEO Strategy:** 630+ prerendered pages across 5 languages.
-  **Knowledge Hub:** 11 topic hubs serving as the brand's entry point.

Unified Ecosystem: Shared Supabase backend, common brand identity, and seamless cross-platform user profiles.

High-Level Architecture: Cross-Platform & Cloud Integration



Mobile Tech Stack: React, Capacitor, and Native Plugins

</> SHARED FRONTEND LAYER



React 18 + TS



Vite 5



TailwindCSS 3



Three.js (3D)



MediaPipe Vision



i18next (5 Langs)

CAPACITOR 7 NATIVE BRIDGE

🍏 IOS NATIVE (SWIFT)

- **Custom Swift Plugins:** HealthKit, Apple Watch (WCSession)
- **WatchKit App:** Real-time HR streaming to WebApp
- **RevenueCat:** Unified IAP & Subscription management
- **ATT & Tenjin:** Privacy-compliant attribution

🤖 ANDROID NATIVE (KOTLIN)

- **Kotlin Modules:** Google Health Connect API
- **BLE Manager:** Polar, Xiaomi, and chest-strap support
- **Foreground Service:** Persistent HR tracking
- **WebView Architecture:** Optimized for Android API 45+

Biometric Core: Real-Time Biofeedback & Sensor Fusion

Apple Ecosystem

Real-time streaming from Apple Watch via **WCSession**. Historical data sync (sleep, mindfulness) through **HealthKit**.

WATCHKIT

HEALTHKIT

Android Ecosystem

Unified health data access via **Google Health Connect**. Foreground service for long-duration session stability.

KOTLIN

HEALTH CONNECT

ONDA BioOS
SENSOR FUSION

BLE Peripherals

Direct integration with Polar, Xiaomi, and other chest-straps or rings via custom **BluetoothManager.kt**.

BLE 5.0

POLAR/XIAOMI

Eye-Scan (CV)

Camera-based Autonomic Nervous System (ANS) detection using **MediaPipe Tasks Vision** for touchless biofeedback.

MEDIAPIPE

COMPUTER VISION

Scalable Backend: Supabase, PostgreSQL, and Edge Functions

PostgreSQL & Security

 **Schema Evolution:** 20+ managed migrations ensuring consistent database state across environments.

 **Core Entities:** Optimized tables for `user_profiles`, `practice_rewards`, and `user_progress`.

 **RLS Policies:** Row-Level Security enforced at the database level, isolating user data by `auth.uid()`.

Serverless Edge Layer

 **analyze-emotion:** Deno-based function leveraging OpenAI GPT-4 for real-time emotional sentiment analysis.

 **revenuecat-webhook:** Automated lifecycle management for trials, renewals, and churn events.

 **-delete-account:** GDPR-compliant cascading deletion of user profiles and authentication records.

 **Storage:** Audio bucket for 5-30 min practices with TUS resumable uploads and CDN-edge caching.

Content Engine: 630+ Prerendered Pages & SEO Strategy

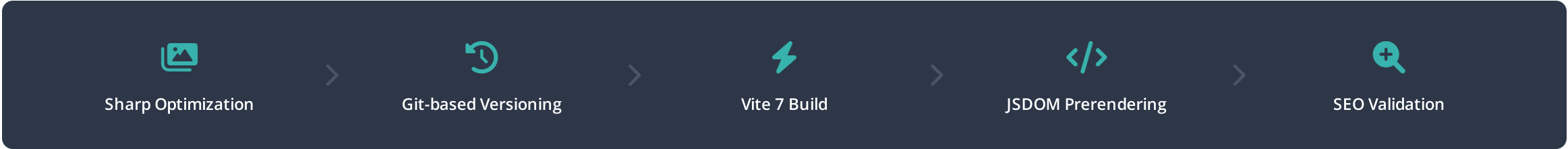
68
ARTICLES

216
GLOSSARY TERMS

11
TOPIC HUBS

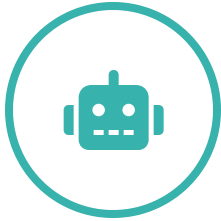
630+
PRERENDERED URLS

Static Site Generation (SSG) Pipeline



- React 19
- TypeScript 5.9
- TailwindCSS 4
- i18next (5 Languages)
- Express + Helmet

Technical Roadmap: AI Personalization & Advanced Biofeedback



AI Personalization Engine

GPT-4 based emotional analysis integration.
Dynamic session adjustment based on real-time HRV.



Advanced Biometric Fusion

MediaPipe eye-scan for ANS detection.
Multi-sensor synchronization (Watch + BLE + Camera).



Predictive Health Modeling

Longitudinal data analysis for burnout prediction.
Establishing clinical standards for digital interventions.

Building the Future of Bio-Responsive Digital Health

ONDA LIFE

Biological Systems & Neuro-Physiological Framework

SCIENTIFIC FRAMEWORK FOR RESEARCH LABORATORIES • 2026

Evolutionary Architecture: From Cellular Zero to Quantum Consciousness

BLOCK 1	Biological Zero & Locomotion	Levels 1-6: Amoeba, Fish, Amphibian, Pack	
BLOCK 2	Cognitive & Social Architecture	Levels 7-12: Clarity, Focus, Expression, Co-creation	
BLOCK 3	Neuro-Somatic Mastery	Levels 13-18: Somatic Intelligence, Network Sync	
BLOCK 4	Legacy & Bio-Design	Levels 19-21: Legacy Code, Bio-Design, Synthesis	
BLOCK 5	Quantum Singularity	Levels 22-24: Resonance, Totality, The Source	

Levels 1-6: Biological Zero & Primary Locomotion

⚙️ Levels 1-3: The Foundation

1. **I Am (Amoeba):** Cellular homeostasis, primary interoception (Insula), and the "Biological Zero."
2. **I Move (Fish):** Activation of Central Pattern Generators (CPG) and axial spinal dynamics.
3. **I Adapt (Amphibian):** Mastering gravity via Vestibular-Ocular Reflex (VOR) and proprioceptive loops.

HOMEOSTASIS

CPG

VOR

👥 Levels 4-6: Space & Social

4. **I Maneuver:** Navigating 3D space and environmental interaction.
5. **I Guard:** Territorial awareness and the regulation of defensive neural circuits.
6. **I am in a Pack:** Transition to social safety and ventral vagal engagement.

SPATIAL NAVIGATION

VENTRAL VAGAL

Research Focus: Validating the transition from "survival mode" to "social safety" via HRV and cortisol modulation.

Levels 7-12: Cognitive Architecture & Social Co-Creation

💡 Cognitive Mastery (7-9)

7. **I Discern:** Developing clarity and the ability to filter essential signals from noise.

8. **I Focus:** Sustained attention and the heart of cognitive endurance.

9. **I Create Image:** Launching the architect's mind; mental modeling and visualization.

PREFRONTAL CORTEX

DMN REGULATION

🤝 Social Co-Creation (10-12)

10. **I Express:** Authentic self-expression and social communication.

11. **I Interact:** Mastering social dynamics and interpersonal resonance.

12. **I Co-create:** The culmination of social intelligence; collaborative creation.

MIRROR NEURONS

COLLECTIVE INTELLIGENCE

Research Opportunity: Investigating the impact of ONDA's cognitive protocols on [Attention Span](#) and [Social-Neural Synchronization](#) during collaborative tasks.

Levels 13-18: Neuro-Somatic Mastery & Network Sync

⚡ Levels 13-15: Somatic Intelligence

- 13. **I Feel (Neuro-Somatic):** Awakening the neuro-somatic mind and deep sensory intelligence.
- 14. **I Channel:** The body as a high-fidelity conductor for neural and energetic signals.
- 15. **I Connect:** Neuro-somatic fusion; the transition from active management to unified presence.

SOMATIC INTELLIGENCE

NEURAL CONDUCTIVITY

🔗 Levels 16-18: Network Sync

- 16. **I Observe:** Transition from control to pure observation; the "Witness" state.
- 17. **I Integrate:** Architecting internal connections and cross-system integration.
- 18. **I Synchronize:** The "Networked Human"; achieving full coherence across all neural networks.

NETWORK COHERENCE

WITNESS STATE

Research Focus: Measuring cross-network coherence and the physiological markers of the "Witness" state in advanced practitioners.

Levels 19-21: Legacy Code & Biological Design



I Remember

LEGACY CODE

- ✓ **Ancestral Patterns:** Working with inherited biological programs and "Legacy Code."
- ✓ **Epigenetic Memory:** Identifying and integrating deep-seated physiological narratives.



I Restore

BIOLOGICAL DESIGNER

- ✓ **Active Restoration:** Transitioning from observation to active biological system design.
- ✓ **Systemic Repair:** Conscious intervention in neural and somatic pathways.



I Synthesize

EVOLUTIONARY CREATOR

- ✓ **Path Culmination:** Synthesizing Flow (AQUA), Structure (AER), and Vibration (IGNIS).
- ✓ **Unified Presence:** Achieving a state of indivisible biological and mental presence.

Research Focus: Investigating the neuro-plastic markers of "Biological Design" and the synthesis of multi-modal physiological inputs.

Levels 22-24: Quantum Resonance & The Source

“I” Level 22: I Resonate

Neural Superconductivity: Transition to software-based reality management. Zero mental or physical resistance.

⌘ Level 23: I am Totality

Quantum Point Zero: The culmination of the ONDA system. Dissolution of the observer-observed duality.

LEVEL 24: I AM THE SOURCE

Pure Awareness & Creation: Returning to the "Point Zero" of existence. The state where life no longer "happens" to you, but "issues" from you.

I = Source = Consciousness = Reality

⚡ Gamma-bursts of ultra-high intensity

📶 ~100% Brainwave Coherence

👤 DMN Annihilation (Zero Ego-Noise)

Research Frontiers: Validating the 24-Level Neuro-Physiological Path

A Multi-Dimensional Roadmap for Clinical & Fundamental Neuroscience



Autonomic & Cognitive Stability

Focus on **Blocks 1-2**. Investigating the transition from survival to safety and the optimization of cognitive focus.

HRV (RMSSD)

BASAL CORTISOL

ALPHA-THETA RATIO



Neuro-Somatic Integration

Focus on **Block 3**. Studying the body as a conductor and the synchronization of neural networks.

PROPRIOCEPTIVE ACCURACY

NEURAL CURRENT EFFICIENCY



Evolutionary Bio-Design

Focus on **Block 4**. Validating longitudinal neuroplasticity and the synthesis of evolutionary protocols.

STRUCTURAL INTEGRATION

EPIGENETIC MARKERS



Quantum Consciousness

Focus on **Block 5**. Exploring states of "Pure Awareness" and the annihilation of the Default Mode Network (DMN).

DMN ANNIHILATION

GAMMA-BURST INTENSITY

100% COHERENCE

Collaborative Evolution

JOIN THE ONDA RESEARCH NETWORK

We invite academic institutions and research laboratories to partner with us in validating the 24-level neuro-physiological framework. Together, we can explore the frontiers of human potential, from autonomic stability to quantum consciousness.

www.onda-life.com

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